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Week 4 intro

Lesson 1: Confidence Intervals

Lesson 2: Point Estimation, Part I

Lesson 3: Point Estimation, Part II

Weekly Assignment

Graded Assignment: Week 4 Assignment - Confidence Intervals & Point Estimation

Your grade: 100%

Your best: 100%

Your highest: 100%

Week 4 Assignment - Confidence Intervals & Point Estimation

Problem Set-up

Review Learning Objectives

A particular area contains 8,000 condominium units. In a survey of the occupants with sample size 100, 12% of the respondents said they planned to sell their condos within the next year.

COOCN

Ready to review what you've learned before starting the assignment? I'm here to help.

1. Compute the 95% confidence interval for the proportion of people planning to sell. (Hint: The variance of the Bernoulli distribution with parameter p is given by $p(1 - p)$.)

15 / 15 points

The confidence interval will be given in an ordered pair rounded to the third decimal place, such as [0.000, 0.000].

What is the lower bound of the confidence interval?

0.056

Submitted Nov 2, 12:57 PM EDT

Try again

Correct

$$\hat{p} \pm 1.96 * \sqrt{\hat{p} * (1 - \hat{p}) / (n - 1)}$$

Your grade: $\hat{p} = 0.12$
To pass you need at least 75% of your highest score.

100%

View submission

See feedback

2. What is the upper bound of the confidence interval?

0.184

Correct

$$\hat{p} \pm 1.96 * \sqrt{\hat{p} * (1 - \hat{p}) / (n - 1)}$$

Your grade: $\hat{p} = 0.12$
To pass you need at least 75% of your highest score.

Problem Set-up

3. What is the standard error of this estimate? Please answer with at least two significant figures.

0.027

Correct

Problem Set-up

4. What is the variance of $\hat{d}$? Include 2 significant figures in your answer.

0.0018

Correct

Because the two samples are done independently:
$$Var(\hat{d}) = Var(\hat{p1} - \hat{p2}) = Var(\hat{p1}) + Var(\hat{p2}) = s_{p1}^2 + s_{p2}^2$$

5. What is the standard error of $\hat{d}$? Include 2 significant figures in the answer.

0.042

Correct

$$se = \sqrt{Var(\hat{d})}$$

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